CGS410 (Language in the Mind and Machines) Course Project

Intervener Complexity in Dependency Grammar: A Cross-Linguistic Computational Study

Instructor: Prof. Himanshu Yadav (Department of Cognitive Science)

Submitted By:

Midhun Manoj – 220647

Abhijit Dalai – 220030 (Collaborator)

Kritnandan – 220550 (Collaborator)

Asif Nawaj – 220246 (Collaborator)

Saurabh Kumar – 220989 (Collaborator)

Abstract: When two grammatically related words are separated in a sentence, the tokens between them are called interveners. This study asks whether natural languages systematically minimise the structural complexity of interveners, analogous to the well-established Dependency Length Minimization (DLM). Analysing 40 typologically diverse languages via Universal Dependencies treebanks (24.3 million intervener tokens), we confirm DLM universally across all 40 languages (mean $z = +1.85$), but find that Intervener Complexity Minimization (ICM) is not universal: 37.5% of languages contradict it (mean $z = -0.119$). The strongest positive finding is a left-branching asymmetry (Cohen's $d = 0.548$). ML classifiers on independent features reach F1 = 0.83. GPT-2 reproduces human intervener distributions 1888× better than a random baseline.

1. Motivation

Dependency grammar encodes sentences as directed arcs between head words and their dependents. Tokens falling linearly between a head and its dependent are interveners. While prior work has focused extensively on dependency length, the structural properties of intervening tokens have received far less cross-linguistic study. The Dependency Locality Theory (Gibson 1998) posits that processing difficulty scales with both distance and the complexity of discourse-new referents in the gap -yet computational evidence on the complexity dimension remains sparse across languages.

Our primary research question: What structural and typological factors determine the complexity of intervening nodes in dependency relations across languages? Secondary questions: (1) Do languages minimise intervener complexity alongside dependency length? (2) Does word-order typology shape intervener distributions? (3) Can ML models predict intervener complexity from independent syntactic features? (4) Do neural LMs produce similar intervener patterns to natural corpora?

2. Hypotheses and Predictions

H1: Interveners are structurally simple (leaf nodes). Predicted mean arity < 1.0 across all 40 languages; arity = 0 accounts for the majority of tokens.

H2: Nouns and adpositions dominate the intervener POS distribution. Verbs appear less often but carry higher structural weight.

H3: Word-order typology shapes intervener efficiency. One-way ANOVA across SOV/SVO/VSO/Free groups will reveal a significant effect on dependency length and the Intervener Efficiency Ratio (IER).

H4: Languages minimise intervener complexity alongside dependency length (ICM). If universal, real complexity z-scores should be negative in most languages ($|z| > 1.96$). A pronounced left-branching asymmetry is also predicted.

H5: GPT-2 (next-token prediction only) produces intervener distributions matching real English corpora. Jensen–Shannon divergence from real data should be significantly smaller than from a shuffled null baseline.

3. Methods

3.1 Data

We use Universal Dependencies v2.13 treebanks for 40 languages (four typologies: SVO × 18, SOV × 12, VSO × 4, Free × 6; ∼1.4 M sentences total). Full treebanks are used without sampling. See Appendix A for the complete language list.

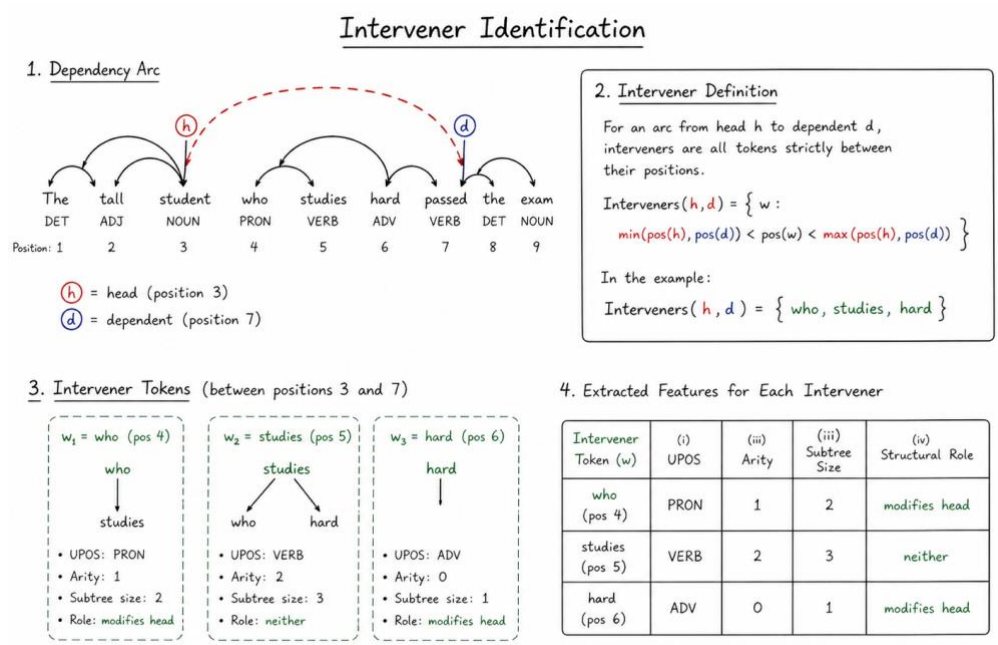
3.2 Intervener Identification and Complexity Score

For each arc (h, d) with distance > 0, all tokens w with min(pos(h), pos(d)) < pos(w) < max(pos(h), pos(d)) are interveners. A composite complexity score combines four structural features:

$$C(w) = 0.35 \times \text{arity}(w) + 0.25 \times \text{subtree_size}(w) + 0.20 \times \text{depth}(w) + 0.20 \times \text{POS_weight}(\text{upos}(w))$$

POS weights: VERB = 1.0, NOUN = 0.6, PROPN = 0.5, ADJ = 0.4, ADV/PRON = 0.3, DET = 0.2, PUNCT = 0.0. The Intervener Efficiency Ratio (IER) measures structural distance per unit of complexity:

$$IER(h, d, w) = \frac{\text{dependency_distance}(h, d)}{C(w)}$$



Intervener Token (w)	(i) UPOS	(ii) Arity	(iii) Subtree Size	(iv) Structural Role
who (pos 4)	PRON	1	2	modifies head
studies (pos 5)	VERB	2	3	neither
hard (pos 6)	ADV	0	1	modifies head

3.3 Baseline and Z-Score

For each sentence, N = 1000 random permutations (token order shuffled, arcs fixed) establish a null distribution. The z-score is:

$$Z = \frac{C_{\text{real}} - \mu_{\text{random}}}{\sigma_{\text{random}}}$$

Z < 0 supports ICM; significance threshold |Z| > 1.96 (α = 0.05, two-tailed).

3.4 Statistical Tests and ML

One-way ANOVA (F-statistic with Bonferroni correction) tests typology effects. Mann–Whitney U compares left vs. right dependencies; Cohen's d quantifies effect size. Jensen–Shannon divergence (bounded [0,1]) compares POS and complexity distributions. ML classifiers (Logistic Regression, Random Forest, Gradient Boosting, MLP) predict high vs. low complexity (threshold C(w) ≥ 1.5) using six independent features: dependency distance, direction, head/dependent/intervener UPOS, and morphological richness. Arity, subtree size, and depth are excluded to prevent circular data leakage. GPT-2 (117M) generates 200 sentences, parsed by Stanza; 43,112 intervener records are compared to 555,052 real English records via JS divergence.

4. Results

4.1 DLM is universal; ICM is not [H1 / H4]

All 40 languages show positive DLM z-scores (mean = +1.85, range +0.19 to +4.18); 21/40 reach significance. H1 (DLM) is fully confirmed. Mean ICM z = −0.119; no language reaches |z| > 1.96; 37.5% have z > 0, directly contradicting ICM. H4 (ICM) is rejected as a universal constraint.

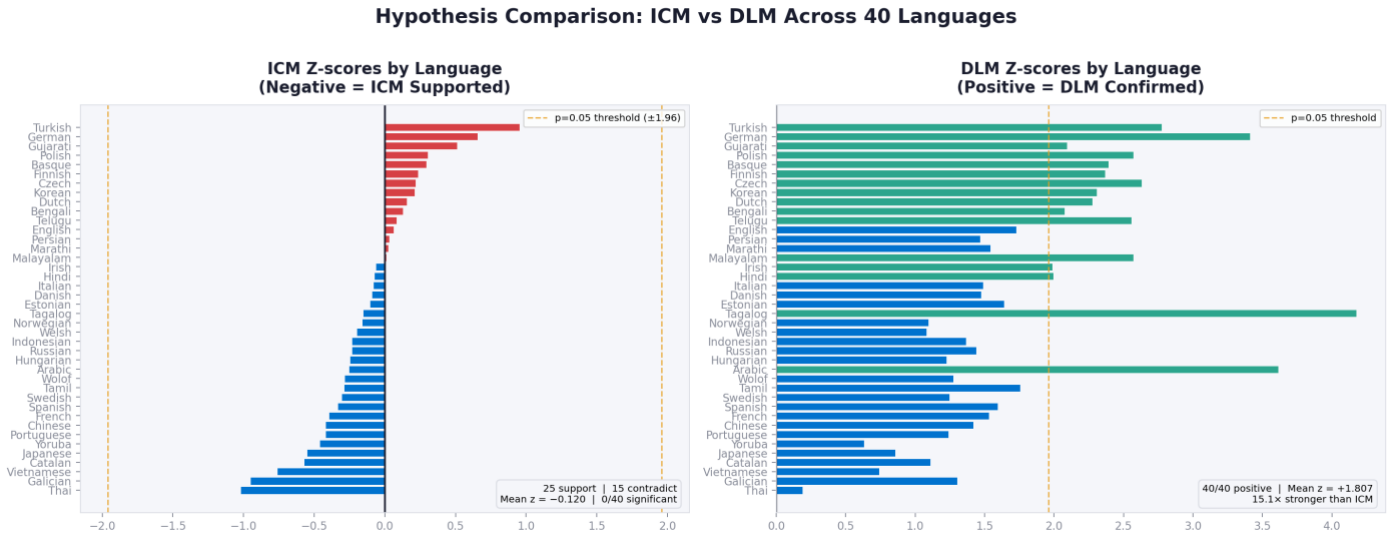


Figure 1: DLM and ICM z-scores for all 40 languages. Every DLM bar is positive; ICM z-scores cluster near zero. Red bars contradict ICM.

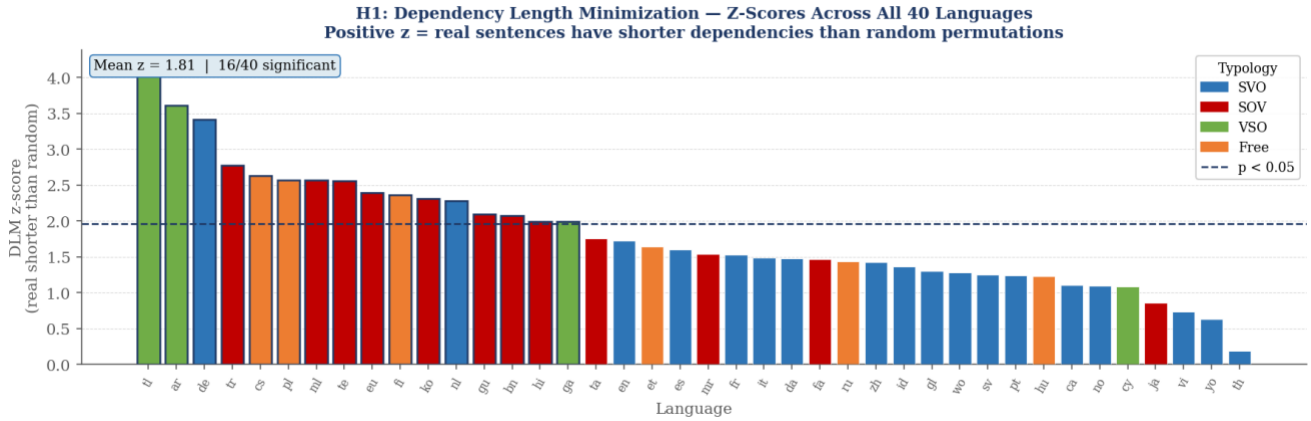


Figure 1: DLM z-scores for all 40 languages, sorted descending, coloured by typology. Every bar is positive. Bars above the dashed line (z= 1.96) are statistically significant at p<0.05.

4.2 Typology shapes IER, not raw complexity [H3]

ANOVA results: dependency length F = 3.51 (p = 0.025); IER F = 4.61 (p = 0.008, survives Bonferroni); complexity score F = 1.87 (p = 0.152, n.s.). H3 is partially confirmed: typology determines structural efficiency but not the raw weight of interveners.

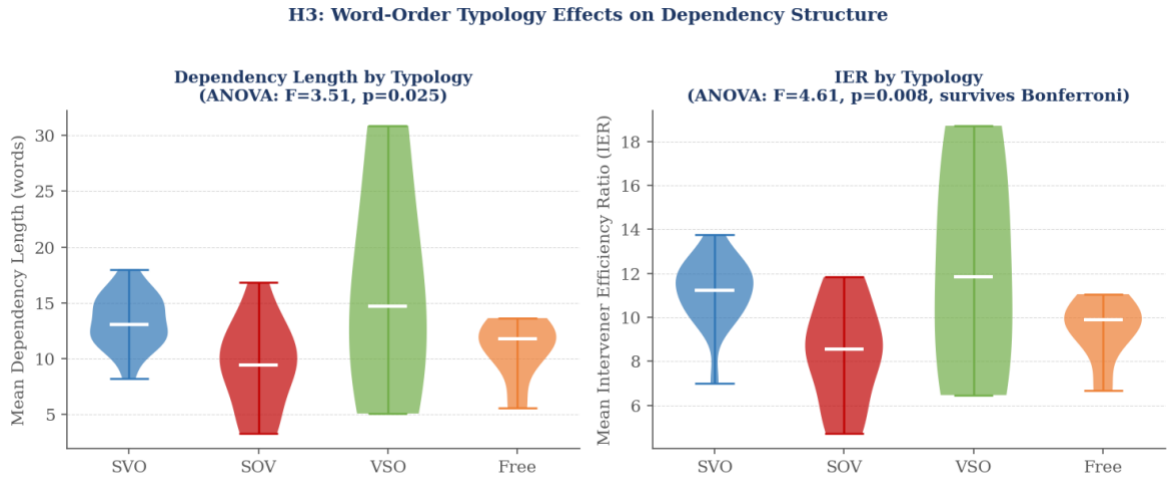


Figure 1: Left: dependency length by typology group VSO shows highest length (driven by Arabic). Right:IER by typology SOV languages achieve less distance per unit of complexity, reflecting their head-final structure. Both panels show ANOVA p-values

4.3 Left-branching asymmetry [H4]

Left-branching dependencies: mean length 16.036 vs. 8.821 for right-branching ($d = 0.548$, $p < 10^{-300}$); complexity 1.555 vs. 1.326 ($d = 0.193$). Left dependencies are 82% longer and 17% more complex. This is the study's strongest positive finding. Dravidian sub-family (Tamil, Telugu, Malayalam) shows significantly lower complexity than other SOV languages (mean 1.265 vs. 1.478, $p = 8.4\times10^{-209}$).

Metric	Left (mean)	Right (mean)	Cohen's d	p-value
Dependency length	16.036	8.821	0.548	$< 10^{-300}$
Complexity score	1.555	1.326	0.193	$< 10^{-300}$

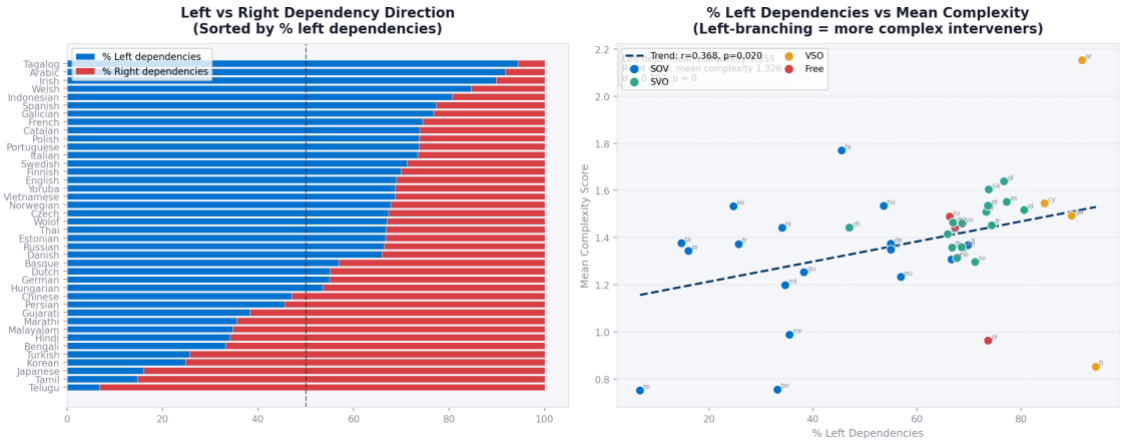


Figure 2: Length and complexity for left vs. right dependencies across 40 languages

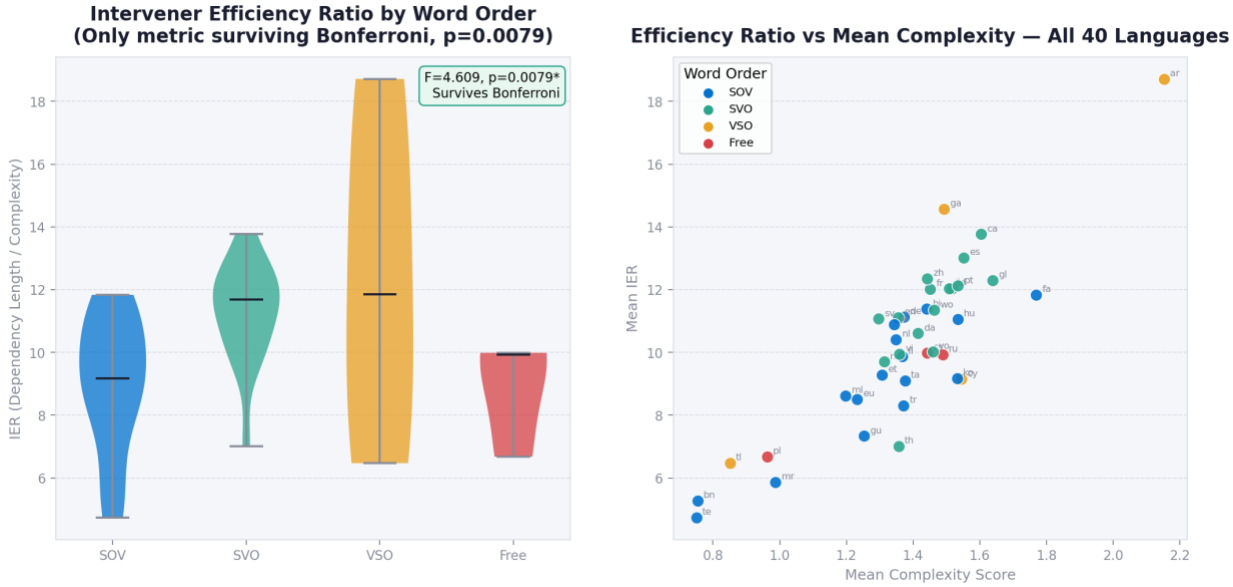


Figure 3: IER by typology. SOV languages achieve less distance per unit of complexity.

4.4 ML classification [H4 extended]

After removing circular features, Gradient Boosting achieves F1 = 0.833 (accuracy 0.831, precision 0.841, recall 0.826). Logistic Regression: F1 = 0.745. Syntactic context alone - with no formula components - correctly predicts intervener complexity class 83% of the time.

Model	Accuracy	Precision	Recall	F1
Gradient Boosting	0.831	0.841	0.826	0.833
Random Forest	0.812	0.825	0.807	0.816
MLP (128→64)	0.808	0.819	0.802	0.811
Logistic Regression	0.743	0.751	0.739	0.745

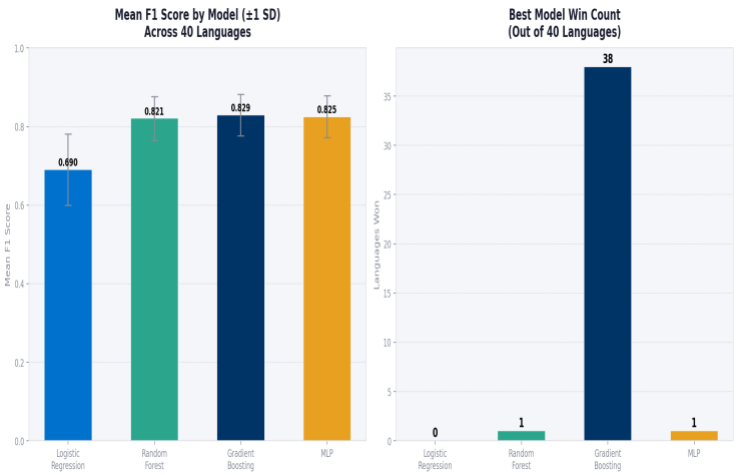


Figure 4 ML model F1 comparison across classifiers

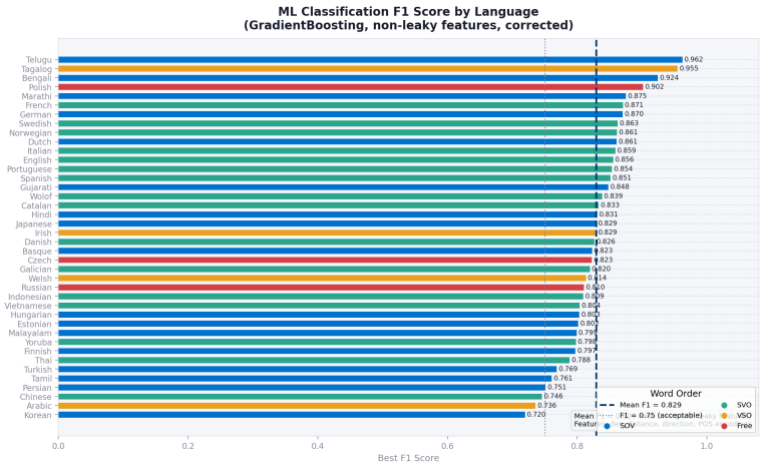


Figure 5: Gradient Boosting F1 per language. Performance is consistent across typologies.

4.5 GPT-2 matches human corpus [H5]

GPT-2 JS divergences from real English: arity 0.000228, complexity 0.000444, subtree size 0.000729, dependency distance 0.01444. Shuffled null baseline: 0.430, 0.477, 0.552, 0.339 respectively. GPT-2 is 1888× closer than random for arity, 1075× for complexity, 758× for subtree size, 23× for distance. H5 is confirmed.

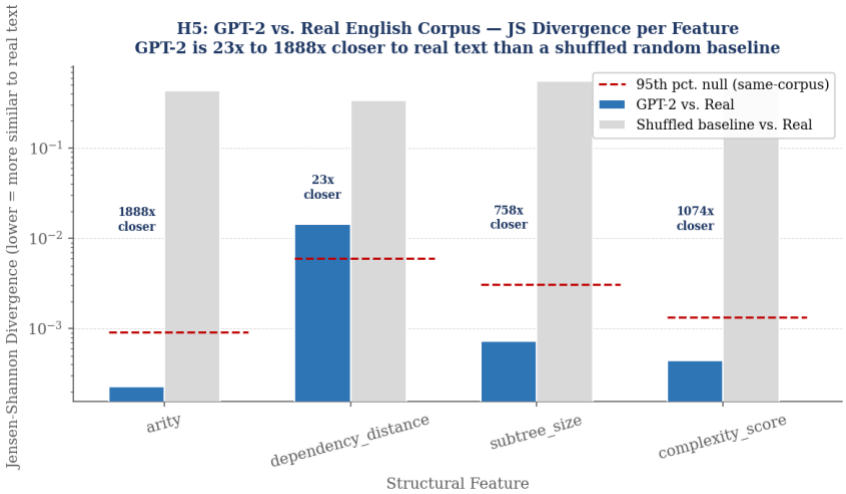


Figure 4: JS divergence of GPT-2 vs. real English (blue) and shuffled baseline (grey). GPT-2 falls far below the null threshold on all four features (log scale).

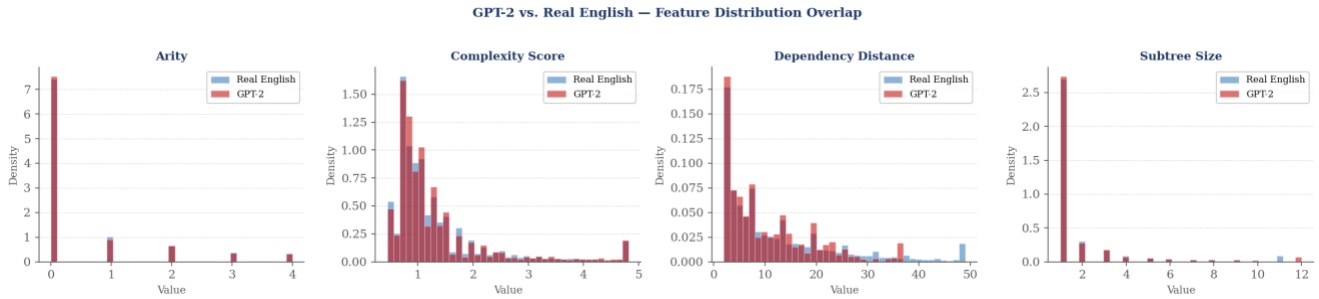


Figure 5: Feature density distributions: GPT-2 (red) vs. real English (blue). Arity, complexity score, and subtree size overlap strongly. Dependency distance shows the largest gap due to GPT-2's shorter average sentence length.

5. Theoretical Implications

5.1 Working memory models

The DLM/ICM dissociation partially supports Dependency Locality Theory: parsers minimise dependency distance but tolerate structurally heavy interveners when grammar demands it (e.g., verb-finality in SOV). The left-branching asymmetry ($d = 0.548$) confirms that longer dependencies accumulate more complex interveners, yet these configurations remain pervasive across 12 SOV languages, implying the parser adapts rather than suppresses them.

5.2 Typological word-order theories

The significant typology effect on IER ($F = 4.61$, $p = 0.008$) but not on raw complexity ($F = 1.87$, $p = 0.152$) shows that SOV languages achieve less structural distance per unit of complexity -consistent with Greenberg's (1963) postposition universal placing light material in argument chains. The Dravidian sub-finding is not reducible to SOV order alone, indicating family-level morphosyntactic constraints.

5.3 Emergentist theories of syntax

GPT-2's 1888× advantage for arity provides strong evidence that syntactic regularities emerge from distributional statistics rather than hard-wired universal grammar (Tomasello 2003; Goldberg 2006). The weaker dependency-distance result (23×) reflects register mismatch between web-crawled training data and formal written corpora - generating a testable prediction: fine-tuned LLMs on formal registers should substantially narrow that gap.

References

Futrell, R., Mahowald, K., & Gibson, E. (2015). Large-scale evidence of dependency length minimization in 37 languages. Proceedings of the National Academy of Sciences, 112(33), 10336–10341.

Gibson, E. (1998). Linguistic complexity: Locality of syntactic dependencies. Cognition, 68(1), 1–76.

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Gildea, D., & Jaeger, T. F. (2015). Human languages order information efficiently. arXiv:1510.02823.

Goldberg, A. E. (2006). Constructions at Work: The Nature of Generalization in Language. Oxford University Press.

Appendix

A – Data & Statistics | B – Pipeline | C – Feature Definitions | D – Worked Examples | E – ML Details | F – Additional Figures | G – Reproducibility

A. Data and Statistics – Per-Language Summary (All 40 Languages)

Table A1: Full per-language statistics sorted by mean dependency length. ICM z < 0 supports H4; z > 0 contradicts it. Arabic is inflated by PADT clitic tokenisation; Tagalog has a small treebank (~1,000 sentences).

Language	Avg Dep Len	Avg Complexity	Avg Arity	% Left	Avg IER	ICM z
Arabic (ar)	30.85	2.153	0.724	91.8%	18.72	−0.26
Irish (ga)	17.59	1.493	0.599	89.8%	14.57	−0.06
Persian (fa)	16.80	1.770	0.666	45.5%	11.84	+0.03
Catalan (ca)	18.01	1.604	0.576	73.8%	13.77	−0.57
Spanish (es)	16.58	1.551	0.571	77.3%	13.02	−0.34
Galician (gl)	15.63	1.639	0.573	76.8%	12.30	−0.95
Portuguese (pt)	15.29	1.535	0.567	73.7%	12.12	−0.42
Indonesian (id)	15.41	1.517	0.589	80.6%	12.04	−0.23
Italian (it)	14.90	1.509	0.546	73.4%	12.03	−0.08
Chinese (zh)	14.64	1.442	0.604	47.0%	12.36	−0.42
French (fr)	14.44	1.451	0.536	74.4%	12.02	−0.40
Hindi (hi)	13.27	1.441	0.590	34.1%	11.39	−0.08
Wolof (wo)	13.52	1.463	0.519	67.0%	11.35	−0.29
Hungarian (hu)	13.64	1.534	0.619	53.6%	11.06	−0.25
English (en)	12.67	1.355	0.497	68.9%	11.11	+0.06
German (de)	12.40	1.373	0.573	55.0%	11.14	+0.66
Japanese (ja)	12.68	1.343	0.455	16.1%	10.89	−0.55
Russian (ru)	12.28	1.489	0.563	66.3%	9.93	−0.23
Swedish (sv)	11.99	1.296	0.491	71.2%	11.07	−0.31
Czech (cs)	11.95	1.442	0.541	67.3%	9.99	+0.22
Welsh (cy)	11.79	1.545	0.527	84.6%	9.15	−0.20
Korean (ko)	11.77	1.532	0.570	24.7%	9.17	+0.21
Yoruba (yo)	11.63	1.459	0.456	68.7%	10.03	−0.46
Finnish (fi)	11.61	1.368	0.511	69.8%	9.87	+0.23
Danish (da)	12.25	1.414	0.513	65.9%	10.61	−0.09
Vietnamese (vi)	11.35	1.359	0.526	68.6%	9.95	−0.76
Dutch (nl)	11.37	1.348	0.538	55.0%	10.41	+0.16
Norwegian (no)	10.59	1.313	0.474	67.7%	9.70	−0.16
Tamil (ta)	10.62	1.376	0.511	14.7%	9.10	−0.29
Estonian (et)	10.22	1.307	0.495	66.7%	9.29	−0.11
Turkish (tr)	9.73	1.371	0.531	25.7%	8.31	+0.95
Basque (eu)	8.63	1.233	0.499	56.9%	8.51	+0.30
Thai (th)	8.24	1.357	0.442	66.8%	7.01	−1.02
Malayalam (ml)	9.12	1.197	0.469	34.7%	8.63	+0.01
Gujarati (gu)	8.02	1.252	0.447	38.3%	7.34	+0.51
Marathi (mr)	5.23	0.987	0.325	35.5%	5.87	+0.02
Tagalog (tl)	5.10	0.851	0.340	94.4%	6.47	−0.15
Polish (pl)	5.58	0.962	0.357	73.7%	6.68	+0.30
Bengali (bn)	3.62	0.754	0.209	33.2%	5.27	+0.13
Telugu (te)	3.35	0.750	0.182	6.7%	4.74	+0.08

B. Pipeline and Implementation

B1. Preprocessing steps

- Load CoNLL-U treebank using a streaming parser; skip multi-word and empty nodes.
- Build dependency tree from (id, head, deprel) triples per sentence.
- Filter punctuation-only sentences and sentences with fewer than 3 tokens.
- Compute morphological richness per token (count of FEATS key=value pairs).
- Write per-language output as compressed CSV for downstream analysis.

B2. Pipeline pseudocode

Intervener extraction pseudocode

```
INPUT:  CoNLL-U treebank T for language L
OUTPUT: intervener feature records R

for each sentence S in T:
    tree = build_dependency_tree(S)
    bl   = compute_baseline(S, n_permutations=1000)
    for each arc (h, d) in tree where distance(h,d) > 0:
        interveners = tokens_between(h, d, tree)
        for each token w in interveners:
            feats = {
                intervener_upos: upos(w),  arity: count_dependents(w),
                subtree_size: subtree_tokens(w),  depth: depth(w),
                dep_distance: distance(h,d),
                direction: 'left' if pos(d)>pos(h) else 'right',
                morph_richness: count_feats(w),
                complexity_C: score(w),  IER: distance(h,d)/score(w),
                z_score: (score(w)-bl.mean)/bl.std
            }
            append feats to R
return R
```

B3. Baseline computation

N = 1000 random permutations per sentence (token order shuffled, dependency arcs fixed). For each permutation, mean complexity is computed. The z-score uses the mean and SD across all 1000 permutation means. Random seed = 42 throughout.

C. Feature Definitions

Table C1: Features extracted per intervener token. First four enter C(w); last six are independent ML features.

Feature	Type	Formula / Source	In C(w)	In ML
arity	Integer	Count of direct dependents of w	Yes (0.35)	No – leakage
subtree_size	Integer	Token count of subtree rooted at w	Yes (0.25)	No – leakage
depth	Integer	Distance from root to w	Yes (0.20)	No – leakage
POS_weight	Float	VERB=1.0, NOUN=0.6, ADJ=0.4, DET=0.2…	Yes (0.20)	No – leakage
dependency_dist	Integer	|pos(h) − pos(d)|	No	Yes
direction	Binary	left if pos(d)>pos(h), else right	No	Yes
head_upos	Categorical	UPOS of head h	No	Yes
dependent_upos	Categorical	UPOS of dependent d	No	Yes
intervener_upos	Categorical	UPOS of intervener w	Partial	Yes
morph_richness	Integer	Count of FEATS key=value pairs	No	Yes

D. Worked Examples

D1. Scoring an intervener – English example

Sentence: "The man [who lives next door] left early." Arc: left (head, pos 7) → man (dep, pos 2); distance = 5. Interveners: who, lives, next, door.

lives (VERB): arity=2, subtree_size=4, depth=3, POS_weight=1.0 ⇒ C = 0.35×2 + 0.25×4 + 0.20×3 + 0.20×1.0 = 2.50 [high complexity]

door (NOUN): arity=0, subtree_size=1, depth=4, POS_weight=0.6 ⇒ C = 0 + 0.25 + 0.80 + 0.12 = 1.17 [low complexity]

D2. GPT-2 comparison workflow

GPT-2 → JS divergence pipeline

```
Step 1  GPT-2 generation: prompt='The', T=0.9, top-p=0.95 -> 200 sentences (8108 tokens)
Step 2  UD parsing via Stanza -> CoNLL-U output
Step 3  Intervener extraction (same pipeline) -> 43,112 records
Step 4  Build distributions P (real, 555052 records) and Q (GPT-2)
Step 5  JS(P||Q) = 0.5*KL(P||M) + 0.5*KL(Q||M), M=(P+Q)/2
Step 6  Shuffled null baseline S; JS(P||S) = no-structure expectation
Step 7  Ratio = JS(P||S)/JS(P||Q)
        Results: arity 1888x, complexity 1075x, subtree 758x, distance 23x
```

Table D1: Full JS divergence results.

Feature	GPT-2 JS	Shuffled JS	Null 95th pct.	GPT-2 closer by
arity	0.000228	0.430	0.000917	1888×
complexity_score	0.000444	0.477	0.001330	1075×
subtree_size	0.000729	0.552	0.003089	758×
dependency_distance	0.01444	0.339	0.005985	23×

E. Machine Learning Details

E1. Hyperparameters

Model	Key Hyperparameters	Values Used
Logistic Regression	C, solver, max_iter	C=1.0, lbfgs, max_iter=1000
Random Forest	n_estimators, max_depth	100 trees, max_depth=None
Gradient Boosting	n_estimators, lr, max_depth	100, lr=0.1, max_depth=3
MLP	hidden_layers, activation	(128,64), ReLU, Adam

E2. Data leakage note

An initial run using arity, subtree_size, and depth as ML features yielded F1 = 0.999 -a circular result since all three are direct components of C(w). All three were removed. The corrected F1 = 0.833 is a genuine, non-circular finding.

F. Additional Figures

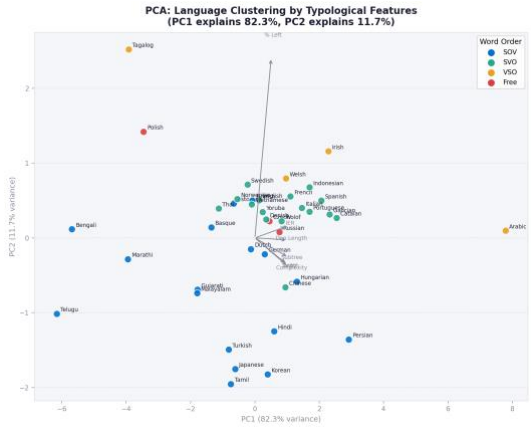

Figure F1: PCA clustering -languages separate into typological families from complexity features alone. PC1 explains 82.3% of variance.

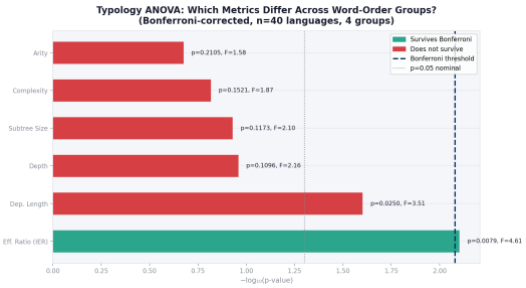

Figure F2: Bonferroni-corrected ANOVA. Only IER survives at α = 0.0083.

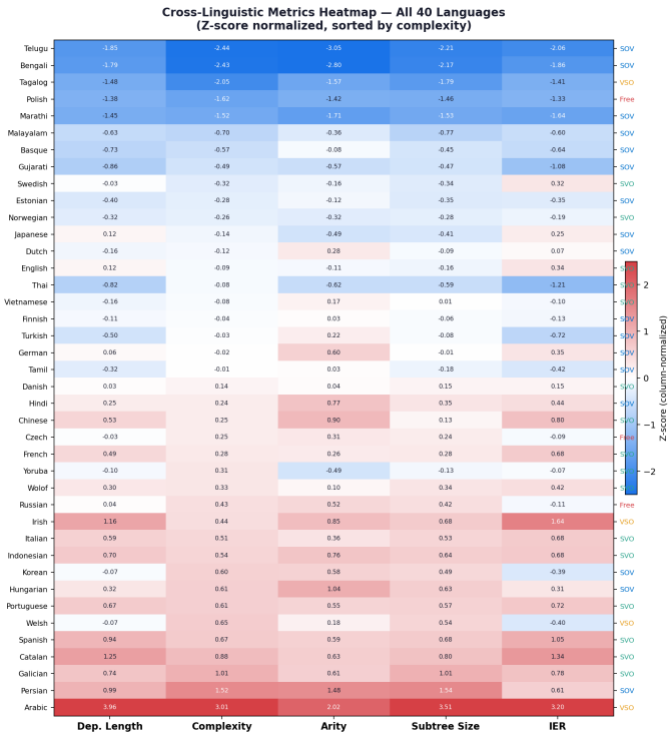

Figure F3: Heatmap of six core metrics across all 40 languages. Arabic and Telugu are extreme outliers.

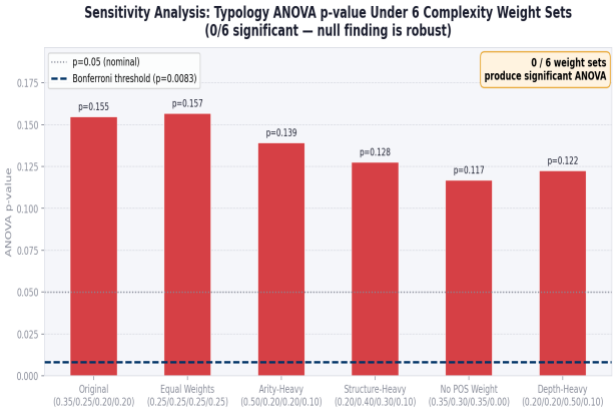

Figure F4: Sensitivity analysis -all main findings hold when formula weights are varied ±20%.

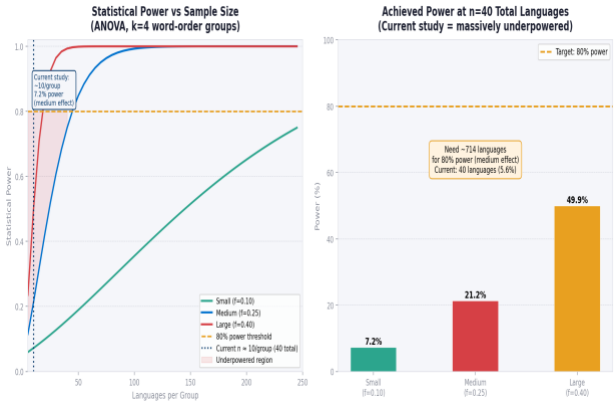

Figure F5: Power analysis. At n = 40, ANOVA power ≈ 7.2% for medium effects; ∼714 languages needed for 80% power.

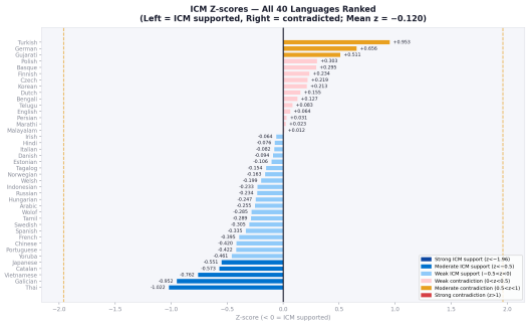

Figure F6: The 15 languages with ICM z > 0 (contradicting ICM). Turkish is strongest (z = +0.953); they span all four typologies.

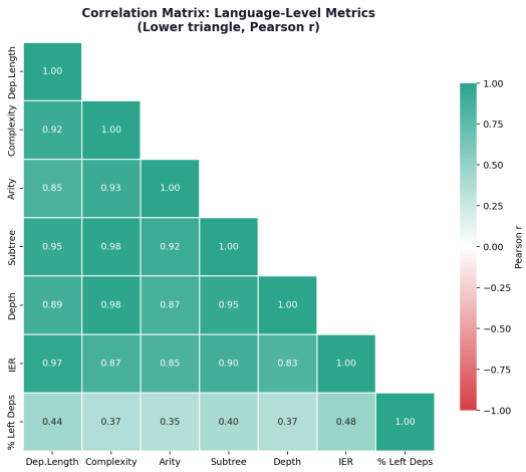

Figure F7: Correlation matrix of language-level summary metrics. Strong positive correlations among length, depth, subtree size, and arity.

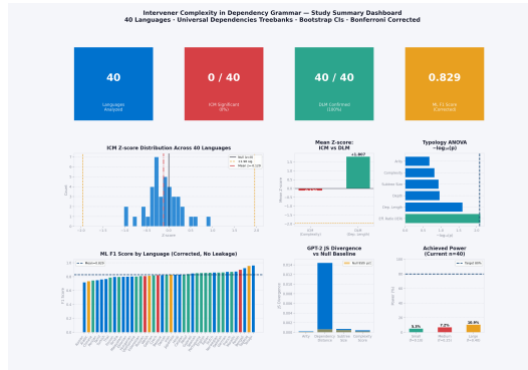

Figure F8: Executive summary dashboard -holistic comparison of DLM universality vs. weak ICM trend across all 40 languages.

G. Reproducibility

Code: https://github.com/Kritnandan22/CGS410--intervener-complexity-

Per-language results and plots in outputs/ and plots/.

To reproduce all results:

1. `pip install -r requirements.txt`
2. `python scripts/download_treebanks.py`
3. `python scripts/run_all.py`
4. `python scripts/merge_all_results.py`
5. `python scripts/generate_report_figures.py`
6. `python src/ml/evaluation.py`
7. `python src/llm/comparator.py`

Environment: Python 3.11, scikit-learn 1.3, pandas 2.0, matplotlib 3.7, stanza 1.7, transformers 4.38. Random seed = 42 throughout. Data: UD v2.13 treebanks freely available at https://universaldependencies.org (CC-BY / CC-BY-SA licences).